

Amazon Sales
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Predicting Product Ratings using Natural Language Processing

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Pipeline

amazon



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Data Overview

product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	rating_count
B07JW9H4J1	Wayona Nyl	Computers&	₹399	₹1,099	64%	4.2	24269
B098NS6PVG	Ambrane Un	Computers&	₹199	₹349	43%	4.0	43994
B096MSW6CT	Source Fas	Computers&	₹199	₹1,899	90%	3.9	7928
B08HDJ86NZ	boAt Deuce	Computers&	₹329	₹699	53%	4.2	94363
B08CF3B7N1	Portronics	Computers&	₹154	₹399	61%	4.2	16905
B08Y1TFSP6	pTron Sole	Computers&	₹149	₹1,000	85%	3.9	24871

about_product	user_id	user_name	review_id	review_title	review_content	img_link	product_link
High Compa Compatible [Fast Cha The boAt D [CHARGE & Fast Charge	AG3D6O4STA AECPFYFQVR AGU3BBQ2V2 AEWAZDZZJL AE3Q6KSUK5 AEQ2YMXSZW	Manav,Adar ArdKn,Nirb Kunal,Hima Omkar dhal rahuls6099 Jayesh,Raj	R3HXWTOLRP RGIQEG07R9 R3J3EQ9TZ R3EEUZKKK9 R18P4L2HH9 R7S8ANNSDP	Satisfied, A Good Bra Good speed Good produ As good as It's prett	looks dura I ordered not quite good produ bought thi its a good	https://m. https://ww https://m. https://m. https://m. https://m. https://m.	https://ww https://ww https://ww https://ww https://ww https://ww https://ww

rating_count	about_product	user_id	user_name	review_id
708 1657 523 1376 121	Customer Care at 959 Resolution: Full HD Accessibility meets USB C to C Cable: Th Resolution: 4K Ultra Color blackLong las	ACVH5GJAFM2JTPYPHRVG AHLHLHYGGLNAPDY7RTZ4N AHVAHTQWBVQ564OYZLFO AE7CFHY23VAJT2FI4NZK AFCWL3MX7BP2ZUDD37ME AHXSEBRVZ06MAYZR606	Amazon Customer,Rahu Mahesh Janardhanan,P Mohd Arshad,Yogesh M Pranav Raj Mohammed, Samson, Rajni Gupta,Lakshmi	R3UKH8PPXQQJ7Q,R1P64 R2XGDUS2ZEQO76,R1GYF R3O19NIP86EJMK,R19RE RUB7U91HVZ30 RC3ZLDRM8CA9T,RMDN4P R1T3IMKX5I23BL,R2ACT

rating_count	about_product	user_id	user_name	review_id
24269 43994 7928 94363 16905 24871	High Compatibility : Compatible with all [Fast Charger& Data The boAt Deuce USB 3 [CHARGE & SYNC FUNCT Fast Charging & Data	AG3D6O4STAQKAY2UVGEU AECPFYFQVRUWC3KGNLJI AGU3BBQ2V2DDAMOAKGFA AEWAZDZZJLQYVVOVG8EU AE3Q6KSUK5P75D5HFYHC AEQ2YMXSZWEOHK2EHTNL	Manav,Adarsh gupta,S ArdKn,Nirbhay kumar, Kunal,Himanshu,viswa Omkar dhale,J.D,HEMAL rahuls6099,Swasat Bo Jayesh,Rajesh k.,Soo	R3HXWTOLRPONMF,R2AJM RGIQEG07R9HS2,R1SMWZ R3J3EQ9TZI5ZJ,R3E7W R3EEUZKKK9J36I,R3HJV R18P4L2HH9TFUP,R16PV R7S8ANNSDPR40,R3CLZF

[www.kaggle.com/datasets/karkavelrajaj/
amazon-sales-dataset](https://www.kaggle.com/datasets/karkavelrajaj/amazon-sales-dataset)

Amazon Dataset 1465 sales transactions

16 column variables
2 character variables
2 numeric variables

Key Variables

Review_ID: R3HXWTOLRPONMF

Review Content: Unique Review

Rating: Range 2 ~ 5

Noticeable factors

Missing values
Random Characters

Motivation & Goals

Goal(s)

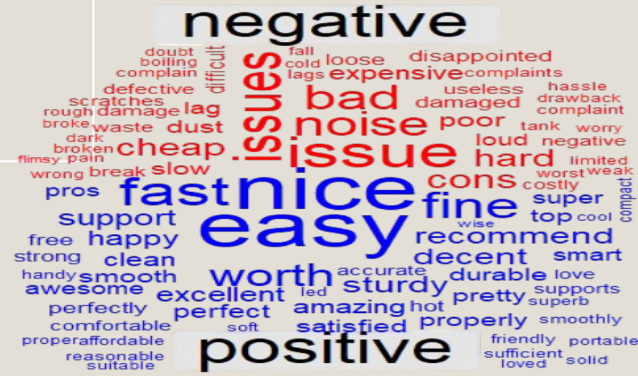
- Predict a 1-5 star rating directly from review text
- Highlight influential terms and phrases
- Compare feature types
 - One-word vs. two-word phrases
- Assess model stability with k-fold cross validation

Relevance/Importance

- E-commerce
 - Reviews shape and influence purchase decisions
 - Early detection of product issues
- Natural Language Processing
 - Raw text → meaningful inputs



EDA & Text Preprocessing



Cleaning

- Remove all missing values (3)
- Remove observations without review ratings
- Clean "review_content" variable
- Remove Duplicates
- Convert to lowercase
- Remove all HTML and numerical test
- Remove punctuation and unnecessary characters
- Removed Stop Words

Tokenization

- Tokenized cleaned "review_content" variable
- Prepare Data for TF-IDF of both Unigram and Bigram

Plot of Bigrams that appear 25 or more times

Unigrams

- Calculate TF-IDF for each tokenized word
- Lacks ambiguity & context

Bigrams

- Calculate TF-IDF for each tokenized 2-word combination
- Preserves some context (ie “bad product” & “great product”)
- Larger chance for sparsity

Document-Term Matrix

- Conversion of both Unigram and Bigram to DTM to prepare for modelling

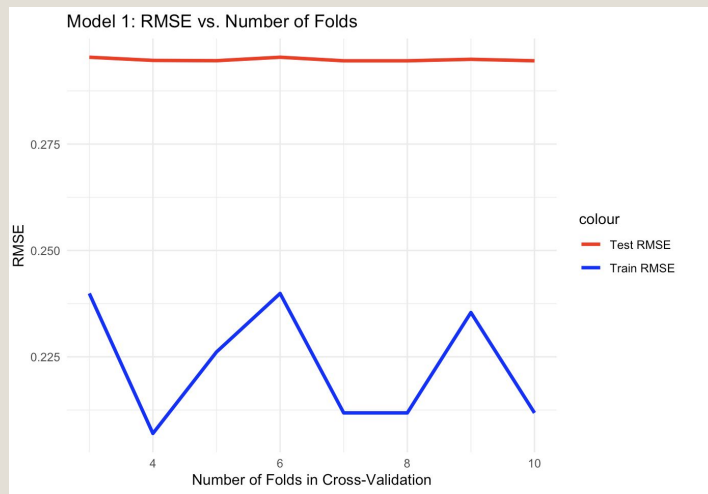
Modelling

Model 1: Lasso Regression with Unigram TF-IDF Features: Predict review ratings using individual words (unigrams) from the cleaned review text.

Approach:

- Each review is tokenized into unigrams (single words).
- TF-IDF is computed to weigh words by their importance in each review and across the dataset.
- A Lasso regression model is trained using these TF-IDF features.

Why Lasso? It performs automatic feature selection by shrinking less useful word features to zero, helping avoid overfitting.



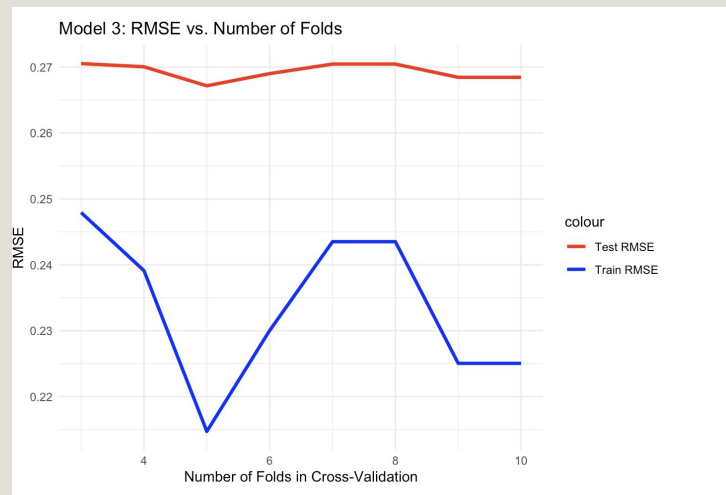
Modelling

Model 3: Lasso Regression with Combined Unigrams + Bigrams:
Improve prediction by using both individual words and 2-word combinations (bigrams).

Approach:

- Builds on Model 1 by also including bigrams, which can capture meaningful phrases (e.g., "not good", "high quality").
- TF-IDF is calculated for both unigrams and bigrams, then combined into one feature matrix.
- Lasso regression is applied to this richer representation.

Advantage: Captures more context from the text, potentially improving accuracy.



Result

Model 1: Lasso Regression (Unigrams Only)

- Captured key single-word signals (e.g., *"excellent"*, *"poor"*)
- RMSE stabilized across folds (3 to 10) showing consistent performance
- Strength: Simplicity and interpretability
- Limitation: Misses important context (e.g., "not bad" ≠ "bad")

Model 3: Combined Unigrams + Bigrams

- Best overall performance
- RMSE reduced further, indicating improved prediction accuracy
- Captures both granular and contextual meaning
- Balanced tradeoff between model complexity and interpretability